Asmodeous (zero): An Atlas of Tiny Predictive Charts Made Real by an Observer

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Abstract

A measurement is the one step of quantum mechanics that is not linear, and it is the step that makes something real. This paper takes that sentence as an engineering principle. The model is an *atlas of many tiny predictive charts*, each with its own small coordinate system, laid over a fragmented geometry. A softmax *observer* plays the part of the measurement: it decides which charts are read, and only the charts it reads are computed ("observed ⇒ exists"). The observer's weights are a partition of unity, so the charts it reads glue into one higher-dimensional space, and the pattern of which charts are read together is a nerve whose homology is the topology the model has learned. The observer is closed into a loop in the manner of a closed timelike curve: it reads the predicted futures of its candidate charts before it commits, and commits to the choice that is consistent with the future it causes. In time, as in space, nothing is computed unless something happens: an event-driven clock fires only on deviation. The mathematics is quantum (linear superposition, then a measurement that breaks linearity) and the machine is a classical consumer GPU. The idea is built as one architecture, ASMODEOUS, assembled slot by slot from the best available part. Its state is a point of $\mathbb{CP}^{N-1}$ that carries its own phase. Its observer is a Parseval-frame POVM whose probabilities sum to one exactly and which includes an outcome for observing nothing. Its charts are gauge-fixed and exist only when read, its gluing maps are learned unitary gauge links, its dynamics are unitary, and its time-traveller is a damped fixed point that provably converges. Its two parents, OBSERVER-ATLAS (real charts under a softmax observer) and PHASE-SPHERE (amplitudes under a Born decider), run beside it as first-class candidates. Every slot has a preregistered ablation, and measurements against owned baselines at equal active FLOPs and equal parameters decide which part earns each claim. **[Results: pending VERDICT.md.]**

Keywords: observer, atlas of charts, partition of unity, lazy computation, Born rule, POVM, Chern number, holonomy, closed timelike curve, event-driven computation, JEPA.

1 Introduction

Computation has a hidden assumption: that the whole state is always there, waiting. A transformer holds every token against every token. A dense network runs every weight on every input. A fixed-rate renderer draws every frame whether or not anything moved. The world does not work like this. A quantum system is a superposition until it is measured, and what is measured is a *choice* of basis that the observer makes. The idea of this paper is to build a learner that is made the same way: **nothing is computed until it is observed, and the observer decides what is worth observing.**

Picture a sphere cut into patches (Figure 4). Each patch is a small model of its own neighbourhood: a tiny predictor with its own few coordinates, valid only where it lives. No single patch covers the sphere, and no single coordinate system ever can; that is a theorem, not a preference. The

sphere is understood only by many small charts that agree where they overlap. Over the patches sits one decider, a softmax. It looks at the situation, assigns weight to the patches, and lets the heavy ones become real. The rest stay as potential: zero cost, zero parameters touched.

That picture carries a chain of ideas, and each link gives the next its reason.

1. **The observer.** Turing's watched-pot paradox, the quantum Zeno effect [5], says that a system measured often enough is frozen. The observer is not a spectator. It is part of the dynamics, and a model of computation that leaves it out leaves out the one operation that decides anything.

2. **Quantum mathematics as an abstraction.** Linear superposition, then a measurement that breaks linearity: that two-step structure runs on a classical GPU as complex linear algebra followed by a normalisation. No quantum hardware and no quantum claim are involved, only the mathematics.

3. **Many tiny charts that become real only when observed.** The geometry is fragmented on purpose. Charts are cheap and numerous; compute is spent only on the few the observer chooses. Parameters scale with the number of charts and compute scales with the number *observed*.

4. **One architecture from the best parts.** ASMODEOUS fills each slot (state, observer, charts, gluing, dynamics, loop, readout, training) with the part that does that job best, whichever parent it comes from. OBSERVER-ATLAS (a real atlas under a softmax observer) and PHASE-SPHERE (a phase sphere under a Born decider, the design of the origin notes) are its parents and run beside it. A part that wins is kept; a part that loses is replaced.

5. **The time-traveller.** The observer is a closed loop. It reads the predicted futures of its candidate charts, and commits only to a choice that is consistent with the future it causes.

6. **Time laziness.** The same principle applies to the clock. Computation fires at events and idles in between, as in the A.E.T.H.E.R. event-driven architecture.

That is the image the name carries. Asmodeus, king of demons, commands a legion, and the legion costs nothing until it is summoned: the king names a few, those few appear and act in their own small corner of the world, and the rest remain names. Here the legion is the atlas of charts, the summons is the observer's softmax, and the corner each one acts in is its chart.

The intended result is a model whose cost follows what matters, whose parameters grow without its compute growing, and whose learned structure can be *read out as a topology* (an integer, a Betti vector, a holonomy) instead of being inferred after the fact. JEPA, mixture-of-experts and attention are vocabulary and baselines in this work, not constraints.

Contributions. The paper builds and tests the following.

- ASMODEOUS: one architecture with a Parseval-frame POVM observer (exact probabilities with $K > N$ outcomes, so laziness costs no approximation), gauge-fixed charts on $\mathbb{CP}^{N-1}$, learned unitary gauge links, unitary dynamics and a damped, provably convergent time-traveller loop (Section 2).

- OBSERVER-ATLAS: an atlas of tiny predictive charts with a gated softmax observer, lazy top-m evaluation, gluing into a higher-dimensional latent space, and an online nerve readout (Sections 4 and 5).

- PHASE-SPHERE: complex amplitudes on a sphere of phases with a Born decider, gauge-fixed charts, and integer readouts (a lattice Chern number and Berry holonomy) (Section 5).
- HYBRID: the composition of the real charts, the coherent decider and the time-traveller loop, built so that each part has a limit in which it vanishes (Section 5).
- A closed-loop observer defined as a fixed point, with an amortised head that predicts its answer in one pass, and an event-driven clock that spends compute only on deviation.
- A suite of owned baselines (single JEPA, MoP-JEPA, chart auto-encoder JEPA, lazy chart auto-encoder JEPA, PEER-JEPA, a CEM planner), matched to the atlas at equal active FLOPs and equal parameters.
- A table of verified identities that the mathematics rests on (Section 6).
- A preregistered measurement protocol, T0–T7 and TO1–TO7, with bars and kills written before any result exists (Section 7).

2 ASMODEOUS: one architecture, the best part in every slot

ASMODEOUS is not a committee of models. Each slot was filled by asking which candidate part does that one job best, and the winner was taken whichever parent it came from: PHASE-SPHERE gives the state, the charts and the topological readouts; OBSERVER-ATLAS gives laziness and gluing; the A.E.T.H.E.R. paper gives the adaptive measurement and the event clock; the CTC loop of the origin notes gives the observer its time-traveller. Every slot has a preregistered ablation that removes it at equal active FLOPs and equal parameters, so a win is credited to a slot only when that slot's ablation loses it.

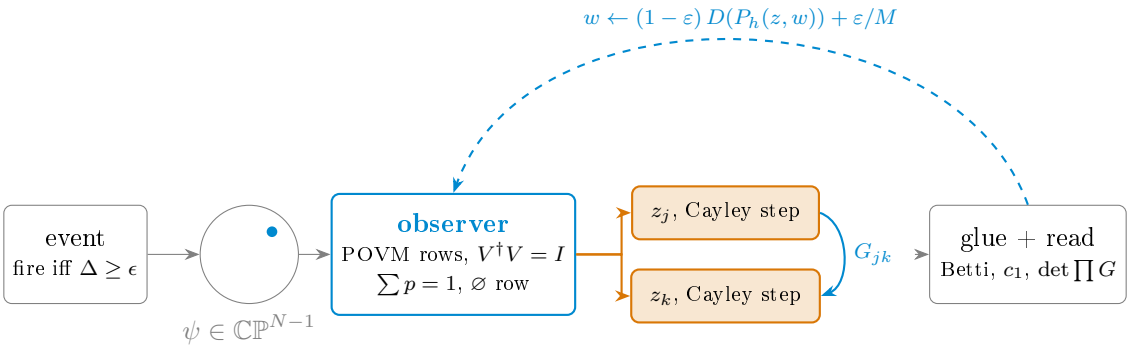


Figure 1: ASMODEOUS, one step. Amber: the charts read this step, the only ones that compute. Blue: the observer, its gauge links and its damped time-traveller loop. Grey: everything that costs nothing until it is called.

The observer. A $K \times N$ isometry with $K > N$ is a tight frame, so its rows already form a POVM with more outcomes than dimensions (Naimark), and every probability is exact:

$$p_k = \left|\langle v_k, \psi\rangle\right|^2, \qquad \sum_{k=0}^{K} p_k = \psi^\dagger V^\dagger V \psi = 1, \qquad \gamma = 1 - p_\varnothing, \tag{1}$$

Slot	Part chosen	From
state	$\psi = \text{normalise}(E(o)) \in \mathbb{CP}^{N-1}$; the phase is part of the state	PHASE-SPHERE
observer	Parseval-frame POVM: rows v_k of an isometry $V \in \mathbb{C}^{(K+1)\times N}$, $V^\dagger V = I$, one row for "observe nothing"	A.E.T.H.E.R., Born, Naimark
charts	$z_k = V_k^\dagger \psi \, e^{-i \arg \alpha_k}$, gauge-fixed, computed only when read	PHASE-SPHERE
laziness in K	score $K{+}1$ effects, run the top-m charts; their weights are exact	OBSERVER-ATLAS
laziness in time	event trigger, adiabatic re-route, per-chart send-on-delta	A.E.T.H.E.R.
gluing	learned gauge links $G_{jk} \in U(d_c)$ with a cocycle loss (a learned Čech cocycle)	origin, PHASE-SPHERE
dynamics	$\hat{z}_k = \text{Cayley}(X_k(a))\, z_k$, unitary	PHASE-SPHERE
observer loop	damped fixed point (a contraction) with an amortised one-shot head	origin CTC
readouts	nerve Betti over $\mathbb{Z}_2$, Chern c_1 from ψ, holonomy $\det \prod G$	both parents
training	fidelity $1 - \lvert\langle \hat{\psi}_{t+1}, \psi_{t+1}^{\text{tgt}} \rangle\rvert^2$; Haar test against collapse	origin, LeJEPA

Table 1: ASMODEOUS slot by slot. The full rationale and the ablation for every row are in `ASMODEOUS.md`.

where v_k is row k of V, $p_\varnothing$ is the probability of the "observe nothing" row and γ is the gate. No partition function appears, so the weights of the top-m read charts are the numbers the dense model would compute: laziness costs no approximation.

The time-traveller. Damping makes the self-consistency map a contraction,

$$w \;\leftarrow\; (1-\varepsilon)\, D\big(P_h(z,w)\big) + \tfrac{\varepsilon}{M}\mathbf{1}, \qquad \varepsilon > 0, \tag{2}$$

so the fixed point $w^\star$ exists, is unique, and is reached geometrically from any start. An amortised head is distilled to produce $w^\star$ in one pass, so the answer is available before the loop that defines it runs.

Training on $\mathbb{CP}^{N-1}$. The prediction loss is the fidelity $1 - \lvert\langle \hat{\psi}_{t+1}, \operatorname{sg} \psi_{t+1}^{\text{tgt}} \rangle\rvert^2$, which is invariant to a global phase and lives on the space the state lives on. Anti-collapse compares sliced $\lvert\langle u, \psi \rangle\rvert^2$ to its Haar law $\text{Beta}(1, N-1)$, the uniform reference on $\mathbb{CP}^{N-1}$.

A control that may take a slot. A single Hopf chart ($\psi \in S^3$ read by the Hopf map to S^2) is preregistered as a control with its own bar. If it covers S^2 without tearing, it becomes the ASMODEOUS chart for that space; a part that wins is kept.

The primitives of Eqs. (1) and (2) are test-bound (Section 6); the full model is built test-first against its contract.

3 Background

3.1 Why an observer?

An observer is the point where a continuum of possibilities becomes one outcome. Quantum mechanics names the operation (measurement), attention names its classical cousin (softmax over

Table 2: Prior art: what each closest owner holds, and what the atlas adds.

Work	What it owns	What the atlas adds
MoP-JEPA [11]	A context-only softmax router over K JEPA predictor heads on a shared encoder, with hard winner-take-all selection; fixes conditional-mean collapse.	A separate chart (encoder and embedding space) for every JEPA; lazy evaluation, so unselected charts cost nothing; gluing of charts into a higher-dimensional space; a topology readout from the observer.
Chart Auto-Encoders [9]	Atlases of local charts with learned encoders, decoders and chart membership, for auto-encoding manifolds.	Predictive (JEPA) charts instead of reconstruction; an observer that routes among charts; lazy top-m evaluation; online nerve topology.
POUnets [6]	Learned partitions of unity that glue local approximations, for regression.	Partition of unity as an observer over latent world-model charts; laziness; a nerve readout of the cover that was learned.
PEER [4]	Retrieval of a few experts out of a very large pool by product keys; parameters grow while compute stays fixed.	Experts that are charts with their own coordinates, glued into one space, with the topology of their co-selection read out.
LeJEPA / SIGReg [1]	An anti-collapse regulariser (SIGReg) for joint-embedding predictive learning.	SIGReg applied per chart and on the glued latent; observer competition as a second, independent anti-collapse mechanism, tested by ablation.
V-JEPA 2 [12]	Action-conditioned latent world models used for planning by model-predictive control.	An observer that reads the predicted futures of candidate charts before committing: a closed-loop decider with a fixed-point condition.
Mixture-of-Depths [7]	Per-token dynamic allocation of compute across layers.	Allocation across charts and across time, with the allocation itself carrying a topological readout.
Mapper [10]	Post-hoc topological summary of a data cloud through a cover and its nerve.	The cover is learned online by the observer, so the nerve is read from the model while it runs.

keys), and the body gives the everyday example: an intention picks a goal, and the geometry of the body computes the path. In every case there is a small decider on top and a rich local mechanism below, and the decider only has to pick, not to compute.

The paradox that opens the project is that *the more often a system is watched, the less it moves.* Rotate a spin by a half-turn in N equal steps and measure after each step; the survival probability of the starting state is

$$P_{\mathrm{surv}}(N) = \cos^{2N}\Big(\frac{\pi}{2N}\Big) \xrightarrow{N\to\infty} 1, \tag{3}$$

where N is the number of measurements during the rotation and P_{surv} is the probability that every measurement finds the system still in its starting state [5]. The observer changes what the system does. This is the first sign that an observer is a dynamical object and not a readout.

3.2 Prior art

Every part of the design has a closest owner. Table 2 names what each owns and what this work adds on top. Each owner is a baseline, a source of vocabulary, and a point of comparison.

4 Mathematical foundation

Every object below is implemented in the `atlas/` package of the accompanying code, and every identity is checked by a reference script and the test suite (Section 6).

4.1 Superposition, then a measurement that breaks linearity

A state is a unit vector ψ expanded in a basis $\{u_k\}$. Evolution is linear. The *decider* is the Born rule, the one non-linear step:

$$|\psi\rangle = \sum_k \alpha_k \, |u_k\rangle, \qquad \alpha_k = \langle u_k|\psi\rangle, \tag{4}$$

$$p_k = \frac{|\alpha_k|^2}{\sum_j |\alpha_j|^2} = \operatorname{softmax}\big(2\log|\alpha_k|\big)_k. \tag{5}$$

Here α_k is the amplitude of ψ along the chart gauge u_k and p_k is the probability of observing chart k.

Proposition 1 (Born is a softmax over log-magnitudes)**.** *For amplitudes $\alpha_k \neq 0$, $|\alpha_k|^2 / \sum_j |\alpha_j|^2 = \operatorname{softmax}(2\log|\alpha_k|)_k$.*

Proof. $|\alpha_k|^2 = \exp(2\log|\alpha_k|)$, and a softmax is the exponential normalised by its sum. □

The second line of (5) is therefore an exact identity. Softmax attention is a classical observer; the Born rule is its coherent form, and the two meet at one formula.

4.2 Measurement as a POVM

A measurement that can be rotated, and therefore learned, is a positive operator-valued measure:

$$E_k = U^\dagger \Pi_k \, U, \qquad \Pi_k = |k\rangle\langle k|, \qquad \sum_k E_k = U^\dagger \Big(\sum_k \Pi_k \Big) U = I, \qquad E_k \succeq 0, \tag{6}$$

where U is a unitary with learnable parameters, Π_k is the fixed projector onto basis state k, and E_k is the effect that selects outcome k. Completeness $\sum_k E_k = I$ holds for every value of the parameters, so the measurement stays valid while it learns, and $p_k = \langle\psi|E_k|\psi\rangle$. The unitary is parametrised by a Cayley transform of a skew-Hermitian generator X, $U = \operatorname{Cayley}(X - X^\dagger)$, which is unitary for every X.

4.3 The softmax observer is a partition of unity; charts glue into $\mathbb{R}^{D_g}$

The observer scores every chart against a query and normalises:

$$w_k(x) = \operatorname{softmax}_k\!\left(\frac{q(x) \cdot \kappa_k}{\sqrt{d}} \right), \qquad w_k(x) \geq 0, \qquad \sum_k w_k(x) = 1, \tag{7}$$

where $q(x)$ is the query computed from the context at input x, κ_k is the learned key of chart k, and d is the key width. The weights sum to one at every input, which is the definition of a partition of unity subordinate to the cover by chart domains. Keep the heaviest m charts $S = \text{top-}m(w)$ and

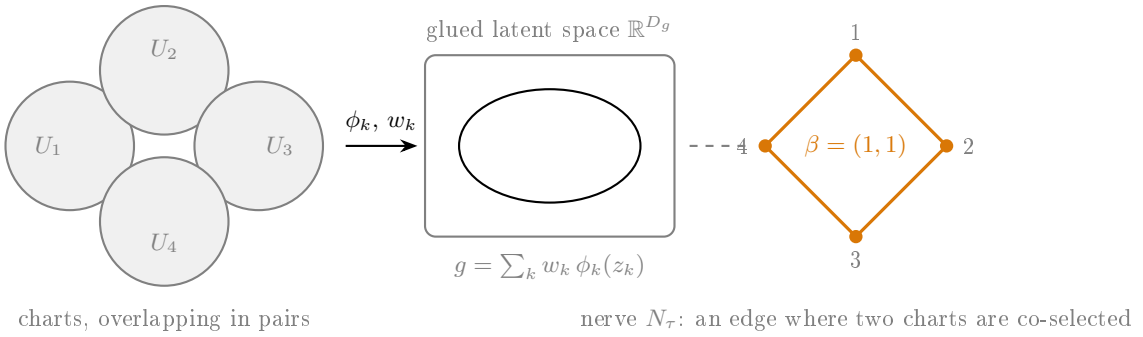


Figure 2: Charts are glued by a partition of unity into the shared latent space $\mathbb{R}^{D_g}$, and the pattern of co-selected charts is a nerve. Four charts overlapping in pairs give a four-cycle, whose Betti vector over $\mathbb{Z}_2$ is $(1, 1)$.

renormalise. Chart k encodes the observation into its own small coordinates $z_k = E_k(o) \in \mathbb{R}^{d_c}$, and a gluing map $\phi_k : \mathbb{R}^{d_c} \to \mathbb{R}^{D_g}$ with $D_g > d_c$ places it in the shared space:

$$g = \gamma \sum_{k \in S} \tilde{w}_k\, \phi_k(z_k), \qquad \tilde{w}_k = \frac{w_k}{\sum_{j \in S} w_j}, \tag{8}$$

where $\gamma \in [0, 1]$ is a learned gate that lets the observer observe nothing (a causal softmax with no gate collapses onto a single token; see Section 6), and g is the glued latent. Predictions glue the same way:

$$\hat{g}_{t+1} = \gamma \sum_{k \in S} \tilde{w}_k\, \phi_k\big(P_k(z_k, a_t)\big), \tag{9}$$

with P_k the tiny predictor of chart k and a_t the action. Figure 2 shows the construction and the nerve that results.

4.4 The lazy rule: observed ⇒ exists

Only the charts in S run their encoder and predictor. Unselected charts are never read. The cost of a step is routing plus m tiny charts:

$$C_{\text{step}} = C_{\text{route}} + m\, C_{\text{chart}}, \qquad C_{\text{route}} = O(K\, d) \text{ (flat keys)} \quad \text{or} \quad O(\sqrt{K}\, d) \text{ (product keys)}, \tag{10}$$

where K is the number of charts, $m \ll K$ the number observed, and C_{chart} the cost of one tiny JEPA. The dense reference costs $K\, C_{\text{chart}}$. Parameters grow with K, compute does not. The lazy top-m computation is checked against the dense mixture that zeroes the same weights (parity), and the unselected charts are checked to cost nothing (Section 6).

4.5 How many charts does a manifold need?

No single chart covers a closed manifold without tearing (invariance of domain). With cat the Lusternik–Schnirelmann category and atlas(M) the least number of ball charts that cover M [8],

$$\operatorname{cat}(M) + 1 \ \le\ \operatorname{atlas}(M) \ \le\ n + 1, \qquad n = \dim M. \tag{11}$$

By these bounds the sphere S^n needs 2 charts, the torus T^2 and the Klein bottle need 3, and $SO(3) \cong \mathbb{RP}^3$ needs 4. The atlas stops tearing at $K = \operatorname{atlas}(M)$, and not before. A rotation in $SO(3)$ has no continuous representation in four Euclidean dimensions or fewer, while five or six suffice [13], which is why the charts are glued into a space of higher dimension than their own.

4.6 The nerve of the observer and its Betti numbers

Charts that are read together form a simplicial complex. For a threshold τ,

$$N_\tau = \Big\{\sigma = \{k_0, \dots, k_r\} \ : \ \exists\, x \text{ with } w_{k_i}(x) \geq \tau \ \ \forall i\Big\}, \tag{12}$$

$$\beta_j = \dim_{\mathbb{Z}_2} H_j\big(N_\tau; \mathbb{Z}_2\big). \tag{13}$$

Here the vertices are the charts, a simplex is a set of charts co-selected with weight at least τ on some input, and β_j is the j-th Betti number over $\mathbb{Z}_2$. By the nerve theorem a good cover has the homotopy type of its nerve, so the learned topology can be read from the observer while the model runs, in contrast to a post-hoc cover-and-nerve summary [10]. Over $\mathbb{Z}_2$ the targets are $S^2 : (1, 0, 1)$, $T^2 : (1, 2, 1)$ and $\mathbb{RP}^3 : (1, 1, 1, 1)$. A partition of unity places at most $\lfloor 1/\tau \rfloor$ charts above τ, and a top-m nerve has dimension at most $m - 1$; both bounds are properties of the readout and are accounted for in its design.

4.7 The phase sphere: amplitudes, transition phases, Chern number, holonomy

The origin notes read "jepa as a phase over higher dimensional S^2 sphere; small phases all around the sphere; the decider will be a softmax". In this reading the state is a point of projective Hilbert space, $\psi \in \mathbb{CP}^{N-1}$ (the Bloch sphere $\mathbb{CP}^1 = S^2$ is the first case), and the small phases are the $U(1)$ fibre. Each chart is a local gauge:

$$\alpha_k = \langle u_k | \psi \rangle, \qquad z_k = V_k^\dagger \psi \, e^{-i \arg \alpha_k}, \qquad \hat{z}_k = \text{Cayley}\left(X_k(a) - X_k(a)^\dagger\right) z_k, \tag{14}$$

where $u_k \in \mathbb{C}^N$ is the learned gauge of chart k, V_k a learned frame, the factor $e^{-i \arg \alpha_k}$ fixes the gauge so that g does not depend on the global phase of ψ, and $\hat{z}_k$ is the local unitary step driven by the action a (a Cayley transform of a skew-Hermitian generator is unitary). Neighbouring gauges differ by a *transition phase*:

$$\chi_{jk} = \arg\big(\alpha_j \, \overline{\alpha_k}\big). \tag{15}$$

The topological readout is an integer. For a triangulated sphere of states, take the link variables $U_{ab} = \langle \psi_a | \psi_b \rangle / |\langle \psi_a | \psi_b \rangle|$ and the plaquette field strength $F_\square = \arg\big(U_{12} U_{23} U_{31}\big)$ on the principal branch. The lattice *Chern number* of Fukui, Hatsugai and Suzuki [3] is

$$c_1 = \frac{1}{2\pi} \sum_{\square} F_\square, \tag{16}$$

where the sum runs over the oriented triangles of the lattice and $c_1 \in \mathbb{Z}$ up to an orientation convention. Its magnitude is 1 for the Hopf bundle $S^3 \to S^2$, the spin-$\frac{1}{2}$ monopole, and 0 for a trivial bundle. Along a closed path the same links give the *Berry phase*, the holonomy of the state:

$$\gamma(C) = -\arg \prod_j \langle \psi_j | \psi_{j+1} \rangle, \qquad \gamma = -\tfrac{1}{2}\, \Omega \ \ (\text{spin-}\tfrac{1}{2}), \tag{17}$$

where C is the closed loop of states and Ω is the solid angle the loop encloses on the sphere. A transported frame returns with a sign ± 1 around the non-orientable loop of a Klein bottle and with sign $+1$ around every loop of a torus, so the holonomy sign separates the two surfaces at the level of individual loops.

4.8 The closed loop: the observer as a fixed point

The loop in the origin notes, the observer's closed timelike curve, is a self-consistency condition (Deutsch's condition $x = F(x)$ [2]):

$$w^\star = D\big(P_h(z,\, w^\star)\big), \qquad w^{(i+1)} = D\big(P_h(z,\, w^{(i)})\big), \qquad \text{halt when } \|w^{(i+1)} - w^{(i)}\| < \varepsilon, \tag{18}$$

where z is the current state, w the observer's choice of charts, $P_h(z, w)$ the predicted state h steps ahead when the charts w are read, and D the decider that maps a predicted future to a choice. The observer's choice must equal the choice it would make after seeing the future that choice causes. A continuous map of the simplex into itself always has a fixed point (Brouwer), so the loop always has an answer and there is no paradox. The Gödel figure of the origin shows the same shape: the halting answer returns to the axis *before* the start event (Figure 6). An *amortised head* learns $\hat{w}(z) \approx w^\star(z)$ in one shot (trained by a KL divergence to the converged $w^\star$), so the answer exists before the iteration that defines it runs.

4.9 The butterfly bonded by topology

The continuous state may be chaotic while the integer readout stays put. With J_t the Jacobian of the latent rollout at step t,

$$\lambda_1 = \frac{1}{h} \log \Big\| \prod_{t=1}^{h} J_t \Big\| \tag{19}$$

is the finite-time largest Lyapunov exponent over the horizon h. The claim is that $\lambda_1 > 0$ (the continuous dynamics is sensitive) while the discrete readouts (c_1, the holonomy class, the Betti vector) cannot change under small perturbations unless a gap closes. The butterfly is bonded by topology.

4.10 Event-driven time laziness (A.E.T.H.E.R.)

The A.E.T.H.E.R. event-driven architecture renders a frame only when the scene has changed enough. With $\mu(t) \in \mathbb{R}^d$ the scene-feature vector, the deviation and the trigger rule are

$$\Delta(t) = \big\|\mu(t) - \mu(t - \delta t)\big\|_2, \qquad \text{fire iff } \ \Delta(t) \geq \epsilon, \tag{20}$$

and the effective rate and the number of frames over a duration T are

$$\mathrm{FPS}(t) \approx \frac{\|\dot{\mu}(t)\|}{\epsilon(t)}, \qquad N_f = \int_0^T \mathrm{FPS}(t)\, dt \approx \int_0^T \frac{\|\dot{\mu}(t)\|}{\epsilon(t)}\, dt, \tag{21}$$

where $\epsilon(t)$ is a threshold that the controller adapts, so the system raises it in noise and lowers it when motion matters. The same architecture defines the adaptive measurement as a POVM $E_k(\Phi) = U^\dagger(\Phi)\, \Pi_k\, U(\Phi)$ with $\sum_k E_k = I$ for every parameter Φ, which is exactly the measurement of Equation (6). The two laziness axes compose: space decides *which charts* run (Equation (10)) and time decides *at which instants* anything runs at all (Figure 3).

5 Method

ASMODEOUS (Section 2) takes its parts from two parents, and both parents also run in full as first-class candidates on identical beds, seeds and metrics through one contract. A part that wins is kept and built on; a part that loses is replaced.

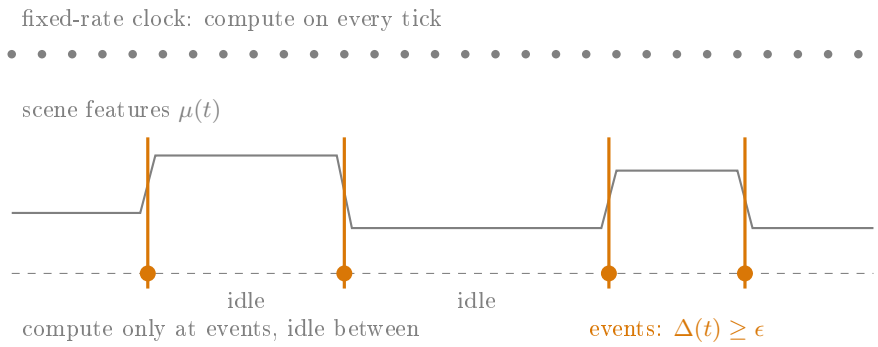


Figure 3: Event-driven time. A fixed-rate clock computes on every tick; the event-driven clock fires only when the scene deviation $\Delta(t)$ reaches the threshold ϵ (accent) and idles in between.

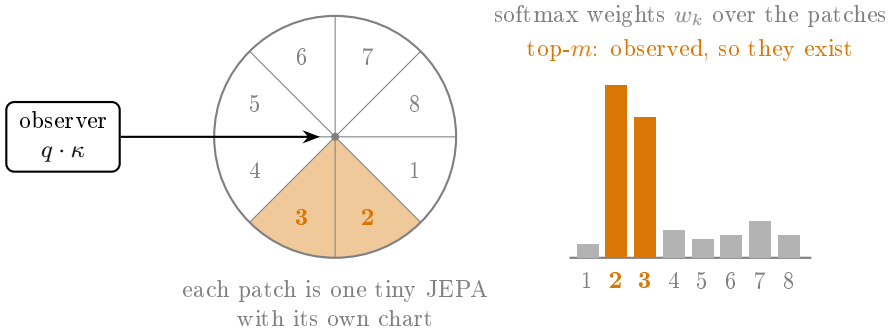


Figure 4: The observer chooses charts on a sphere. A softmax over the patches assigns weights w_k; the heaviest m (here $m = 2$, shown in the accent colour) are observed and therefore computed. The bar heights are illustrative.

5.1 OBSERVER-ATLAS

The atlas is K tiny JEPAs with private charts (Figure 4). Chart k has an encoder $E_k : o_t \to z_k \in \mathbb{R}^{d_c}$ with d_c small, a predictor $P_k : (z_k, a_t) \to \hat{z}_k$ in the same chart, a stop-gradient or EMA target encoder, and a gluing map ϕ_k. A small causal backbone produces the context c_t, the observer turns it into the weights of Equation (7), a gate lets it observe nothing, and only the top-m charts run (Figure 5). Charts are glued by Equations (8) and (9).

The loss is the sum of: a per-chart JEPA loss against the stop-gradient target in the same chart, an overlap consistency term that makes co-selected charts agree in $\mathbb{R}^{D_g}$, a per-chart and glued-latent SIGReg term [1], a router load-balance term, and the task loss through g. Every structure the model uses (charts, routing, gluing, the manifold itself) is learned by SGD; hand-given structure scores as "representable only", never as "works". Anti-collapse is mandatory in every arm, because every self-supervised term here is minimised by a constant; the claim that observer competition alone limits collapse is tested only by ablation.

5.2 PHASE-SPHERE

The design of the origin notes is built as follows. The state is $\psi_t = \text{normalise}(E(o_t)) \in \mathbb{C}^N$. There are K learned chart gauges u_k and a null gauge $u_\varnothing$. A Born decider (Equation (5)) with a "nothing" outcome sets the gate $\gamma = 1 - p_\varnothing$. Lazy gauge-fixed charts z_k are computed for the top-m only, with a

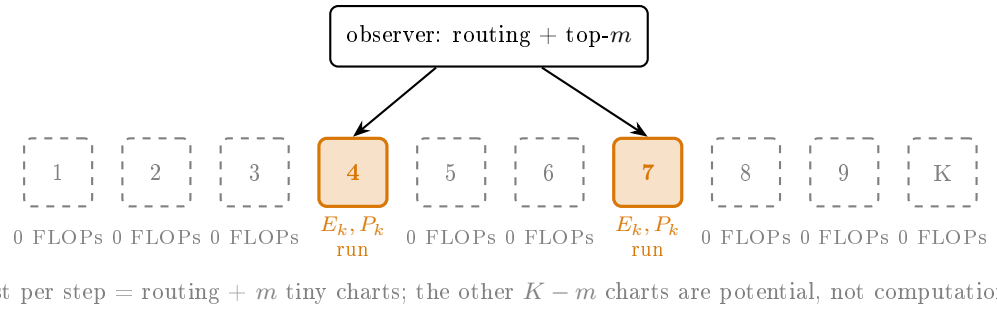


Figure 5: Lazy compute. Only the observed charts (accent) run their encoder E_k and predictor P_k; every unselected chart (dashed grey) costs zero FLOPs.

Cayley-unitary local dynamics (Equation (14)). The glued latent is $g = \gamma \sum_k \tilde{w}_k \, \phi_k([\mathrm{Re}\, z_k, \mathrm{Im}\, z_k])$, the transition phases are χ_{jk} (Equation (15)), and the closed-loop observer of Equation (18) carries an amortised head. Its readouts are the Chern number of Equation (16) and the Berry holonomy along held-out loops. Because amplitudes enter only through $K \cdot N$ Born scores, K needs no product keys.

5.3 HYBRID

HYBRID is the composition of three parts: the real charts of OBSERVER-ATLAS; the coherent decider $p = |U\psi|^2$ with $\psi_k = \sqrt{\mathrm{softmax}(s)_k}\, e^{i\theta_k(q)}$ and U a low-rank Cayley rotation; and the time-traveller loop (START preselects M charts, the life-line rolls their candidate futures, HALT is the self-consistent set). Nothing is bound together by hand: every part has a limit in which it vanishes, and the arm starts at the first of those limits.

$$\begin{aligned} \text{no coherence, no phase} &\Rightarrow \text{OBSERVER-ATLAS}, \\ \text{no loop} &\Rightarrow \text{coherent atlas}, \\ \text{no coherence} &\Rightarrow \text{time-traveller atlas}. \end{aligned} \tag{22}$$

In the implementation the coherence parameters C_p vanish in the no-coherence limit, and a loop gate γ_{h} (distinct from the observer gate of Equation (8)) vanishes in the no-loop limit; the initialisation sits at the first limit, where θ has no effect. The hybrid contains OBSERVER-ATLAS, the coherent atlas and the time-traveller atlas as special cases, and learns how far from each to move. If the fixed-point observer of PHASE-SPHERE replaces the time-traveller loop, that is a recorded change made before any hybrid result, never a post-hoc pick between the two loops.

5.4 The time-traveller

The origin figure is the Gödel-universe closed timelike curve (Figure 6): a life-line leaves the axis at START, spirals out to where the angular circles turn timelike, rides them backward in t, and returns to the axis at HALT, earlier than it left. In the architecture the life-line is the roll-out of candidate futures, and HALT is the self-consistent choice of Equation (18). The observer does not guess a chart and then find out. It reads what each candidate would predict and commits to the choice that survives its own prediction, and an amortised head learns to produce that choice in a single pass.

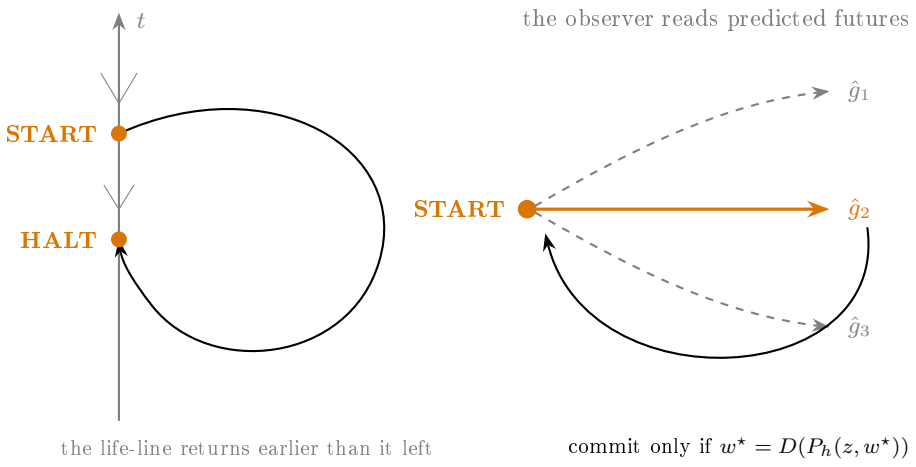


Figure 6: The time-traveller. Left: a closed timelike curve in the Gödel-universe picture; the life-line leaves the axis at START, rides back in t, and returns to the axis at HALT, earlier than it left. Right: the observer reads the predicted futures $\hat{g}_1, \hat{g}_2, \hat{g}_3$ of three candidate charts and commits only to the choice that is a fixed point of its own prediction.

Table 3: The owned baselines and the role each plays.

Arm	Role
Single JEPA, several latent widths	The Whitney baseline: one chart wide enough to embed the manifold.
MoP-JEPA (dense and lazy) [11]	The closest owner: K predictor heads on a shared encoder behind a context-only softmax router.
Chart-AE JEPA [9]	Owner of separate charts and gluing, with membership over all K charts.
Lazy chart-AE JEPA	The observer ablation: the atlas without the backbone, the gate and the keys.
PEER-JEPA [4]	The owned route to "parameters grow, compute stays fixed" by product-key experts.
CEM planner	V-JEPA-2-AC-style [12] model-predictive control over any arm's rollout, with planning compute equalised across arms.

5.5 Owned baselines and the matching rule

Every owned method is a baseline, built to the same contract, trained by SGD, and matched to the atlas at *equal active FLOPs and equal total parameters*. Active FLOPs are the multiply-accumulates per step of the observer's forward pass, and parameters exclude EMA copies; the atlas uses the same counting conventions. Each arm carries the shared SIGReg term and an EMA target, and takes only observations and actions as input. Table 3 lists the arms.

The matching rule (L-BR) sizes each arm by its hidden width against the atlas's active FLOPs and parameters, with a tolerance of $\pm 5\%$ on each axis. Two cases arise. When one configuration of the arm hits both targets, a single baseline runs ("both"). When the arm cannot hold the atlas's ratio of parameters to FLOPs, two baselines run: one at equal compute with fewer parameters ("flop"), and one at equal parameters with more compute ("param"). The atlas must beat each baseline on the axis it is matched on. Dense arms have parameters close to FLOPs, while top-1-of-K arms have a parameter-to-FLOP ratio below K. A "both" match whose smallest width falls below 32 is degenerate (for example a very narrow encoder feeding very wide heads) and is replaced by the

split. Most of the atlas's active FLOPs sit in its causal backbone, and every owned arm spends that budget on encoder width instead.

Hyperparameters are tuned on a separate seed, with the same budget for every arm, the atlas included; widths are fixed by matching and never tuned. The selected configuration is frozen and trained on the evaluation seeds. The HYBRID arm is matched in the same way, with the coherent decider and its extra parameters counted, and its planning budget includes the coherent START decider and the worst-case loop cost.

Where an owned method is reimplemented, the deviations from its source are recorded. MoP-JEPA gains a SIGReg term because anti-collapse is mandatory in every arm (setting its weight to zero restores the paper-native arm, which the collapse test runs). Its unpublished hyperparameters are set by tuning, one prediction is read out at the router's top-1 head, and the latent width is set by matching rather than by the paper. Chart-AE JEPA replaces pixel reconstruction by JEPA losses against an EMA target, and the lazy variant adds top-m routing with a load-balance term so that unselected charts keep a router gradient.

6 Verified identities

Everything the mathematics stands on is a checked identity before it is a design. Every line of Table 4 is produced by the reference script `ref/identities.py` (pinned by SHA-256) or asserted by a passing test in the `tests/` directory of the accompanying code (eight test modules, covering the facts, the phase sphere, the coherent observer, contextuality, the observer atlas, the owned arms, topology, and the systems benchmark).

7 Experiments

Each claim of the project is a question put to a measurement. The beds, the bars and the kills are written into a preregistration file (`PREREG.md`) and committed before any result file exists, and every claim is scored for ASMODEOUS, its two parents and HYBRID under the same beds, seeds, matched sizes and owned baselines. The verdict for a claim names whichever candidate passes it. This section states the questions only; the bars live in `PREREG.md`.

7.1 Measurement discipline

Five rules govern every comparison. (L-GIVEN) Any structure that a bed supplies (charts, routing, gluing, manifolds, graphs) must be learned by SGD, and hand-given structure scores as "representable only". (L-PREREG) Beds, bars and kills are committed before any result exists; anything later is a post-hoc label. (L-BR) Baselines are the best owned methods at equal active FLOPs and equal total parameters, trained by SGD and tuned on a separate seed (Section 5.5). (L-WALD) A designed bed counts only if a neutral bed runs beside it. (L-LEARN) An arm must clear its in-distribution bar before it is compared. A literature gap is not a new primitive: novelty is earned only by a decisive test that the owned parts fail.

7.2 Core tests

T0, reproduce. Does the reference script reproduce its pinned output?

T1, owner hunt. Is the full combination already owned? Every owned part is cited and used as a baseline.

T2, geometry (C1, C2). On the sphere, torus, Klein bottle and $SO(3)$, with no manifold, chart or state ever given, does the atlas stop tearing at exactly $K = \text{atlas}(M)$ with charts of the manifold's own dimension, and does the nerve recover the true Betti numbers online? A neutral flat bed runs beside them.

T3, lazy scaling (C3). At fixed active compute, does loss fall as the number of charts K grows, below a dense JEPA and below one JEPA with PEER experts? Does lazy top-m stay within tolerance of the soft mixture over all K?

T4, collapse (C4). With the anti-collapse regulariser weakened or removed, does observer competition keep the representation from collapsing? Does the atlas match MoP-JEPA on mode collapse on the OGBench pointmaze-stitch task?

T5, time-traveller (C5). Does reading predicted futures before committing beat owned latent MPC at equal planning compute, on MuJoCo swing-up and OGBench pointmaze?

T6, contextuality. Do query pairs show order effects and QQ-equality statistics as a quantum-structured observer predicts? Measurement only.

T7, systems. Do lazy routing and tiny JEPAs run faster on a consumer GPU than a dense JEPA of equal quality?

7.3 Origin-derived readouts

These are scored alongside the core tests, for every candidate where they are defined.

TO1, Chern. Does the learned amplitude bundle read $|c_1| = 1$ on the Berry bed and 0 on its trivial control?

TO2, fixed point. Does the closed-loop observer converge, and does it beat one-pass routing at equal FLOPs?

TO3, halt before start. Does a one-shot head predict the converged $w^\star$ at a fraction of the loop's cost?

TO4, butterfly. Is $\lambda_1 > 0$ while the integer readouts stay put under perturbation?

TO5, scattering front end. Does a multiscale scattering cascade reach the same prediction quality in fewer steps?

TO6, rays through a lattice. Does ray-cast selection over a grid of charts match product keys at lower cost?

TO7, holonomy. Does the sign of the holonomy along held-out loops separate the torus from the Klein bottle?

7.4 The hybrid claim

HYBRID is held to the "best of both" claim directly. On each claim it must be at least as good as the better of OBSERVER-ATLAS and PHASE-SPHERE within that claim's own tolerance, and strictly better than both on at least one preregistered claim, with a paired confidence interval whose lower bound is above zero, at equal matched sizes. The preregistered kill fires if composition costs

more than it adds, or if every learned limit collapses so that the hybrid has learned to be the atlas. Ablations that switch each part off (coherence parameters frozen at zero, the loop gate at zero, the phases frozen at zero) decide which part earned a win, under the same budget; a hybrid win counts only for the part whose ablation loses it. The Chern and holonomy readouts are eval-only and never enter a loss. A real-valued arm has a trivial amplitude bundle by construction, so it reads $c_1 = 0$, which counts as a miss on the Berry bed and not as a void.

8 Results

9 Verdict

10 Conclusion

References

[1] Randall Balestriero and Yann LeCun. LeJEPA: Provable and scalable self-supervised learning without the heuristics, 2025.

[2] David Deutsch. Quantum mechanics near closed timelike lines. *Physical Review D*, 44(10):3197–3217, 1991.

[3] Takahiro Fukui, Yasuhiro Hatsugai, and Hiroshi Suzuki. Chern numbers in discretized Brillouin zone: Efficient method of computing (spin) Hall conductances. *Journal of the Physical Society of Japan*, 74:1674, 2005.

[4] Xu Owen He. Mixture of a million experts, 2024.

[5] B. Misra and E. C. G. Sudarshan. The Zeno's paradox in quantum theory. *Journal of Mathematical Physics*, 18(4):756–763, 1977.

[6] Partition of unity networks: Deep HP-approximation, 2021.

[7] David Raposo et al. Mixture-of-depths: Dynamically allocating compute in transformer-based language models, 2024.

[8] Yuli Rudyak and Felix Schlenk. Minimal atlases of closed symplectic manifolds, 2006.

[9] Stefan Schonsheck, Jie Chen, and Rongjie Lai. Chart auto-encoders for manifold structured data, 2019.

[10] Gurjeet Singh, Facundo Mémoli, and Gunnar Carlsson. Topological methods for the analysis of high dimensional data sets and 3D object recognition. In *Eurographics Symposium on Point-Based Graphics*, 2007.

[11] Zhi Song et al. MoP-JEPA: Hard-assigned predictor mixtures for stochastic JEPA world models, 2026. version 1.

[12] V-JEPA 2: Self-supervised video models enable understanding, prediction and planning, 2025.

[13] Yi Zhou, Connelly Barnes, Jingwan Lu, Jimei Yang, and Hao Li. On the continuity of rotation representations in neural networks, 2019.

Table 4: Verified identities. The first column names the identity, the second its value, the third how it is checked.

Identity	Value	Check
Pin: SHA-256 of `ref/identities.py`	`467c2ae2...40a3fe`	matches the pin
Zeno: $\cos^{2N}(\pi/2N)$, $N = 1, 2, 10, 100, 1000$	0.000000, 0.250000, 0.780546, 0.975627, 0.997536	simulated survival equal to 6 decimals
Born: $p = \text{softmax}(2\log\lvert\alpha\rvert)$	$= \lvert\alpha\rvert^2 / \sum\lvert\alpha\rvert^2$	atol 10^{-12}, 100×9 complex128
Lüders: $\max\lvert QQ\rvert$, pure and mixed states	$< 10^{-14}$	QQ equality holds
Classical: $\text{mean}\lvert QQ\rvert$, with disturbance	0.0751	QQ equality breaks
Softmax: causal stack, mass on token 1	0.2543 ($L = 1$) $\to$ 1.0000 ($L = 16$)	reference script or test suite
IIA: p_1/p_2, key 3 present / removed	softmax 1.3499 / 1.3499; coherent $g = 1$: 2.0587 / 1.3499	reference script or test suite
Cayley: $U^\dagger U = I$	50 random generators	atol 10^{-10}
POVM: $\sum_k E_k = I$, $E_k \succeq 0$	$K = N$ and Naimark $K > N$, 50 draws each	atol 10^{-10} (`tests/test_povm.py`)
Frame: $V^\dagger V = I$, Stiefel retraction	$K \in \{16, 65, 257\}$	$< 10^{-10}$ (`tests/test_asmodeous_ops.py`)
Tight frame: $\sum p = 1$ for $K > N$ outcomes	$p \geq 0$	$< 10^{-10}$
Damped CTC loop: unique fixed point	10 random starts	residual $< 10^{-8}$; undamped cycle does not converge
Gauge links: G_{jk} unitary, exact cocycle	random generators	$< 10^{-10}$
Parity: lazy top-m = dense masked mix	$K = 8$ flat keys, $K = 16$ product keys	atol 10^{-10} (float64)
Laziness	g, $\hat{g}$, loss unchanged (atol 10^{-12}); dense control $\to$ NaN	unselected charts NaN-poisoned
FLOPs: active MACs, m vs K	per-chart cost exact in m; K enters only the router	reference script or test suite
Kernel: fused Triton step = `observe()`	$K = 16 \ldots 4096$, flat and product keys	10^{-4}
Berry: $\gamma = -\Omega/2$, spin-$\frac{1}{2}$ latitude loops	5/5 latitudes; 200 triangles exact to 10^{-9}	reference script or test suite
Chern: FHS lattice, levels $2, 3, 4$	spin-$\frac{1}{2}$ -1; spin-1 -2; trivial 0	reference script or test suite
Holonomy: transported sign	Klein -1; torus $+1$	reference script or test suite
Fixed point: $w^\star = D(P(z, w^\star))$, 1000 stochastic maps	residual $< 10^{-8}$	reference script or test suite
Betti $\mathbb{Z}_2$: S^2, T^2, Klein, $\mathbb{RP}^3$	$(1, 0, 1)$; $(1, 2, 1)$; $(1, 2, 1)$; $(1, 1, 1, 1)$	reference script or test suite
Betti $\mathbb{Q}$: T^2, Klein	$(1, 2, 1)$; $(1, 1, 0)$	reference script or test suite
Nerve: top-m readout, 20 seeds	$\dim \leq m-1$; PoU at τ: $\dim \leq \lfloor 1/\tau \rfloor - 1$	reference script or test suite